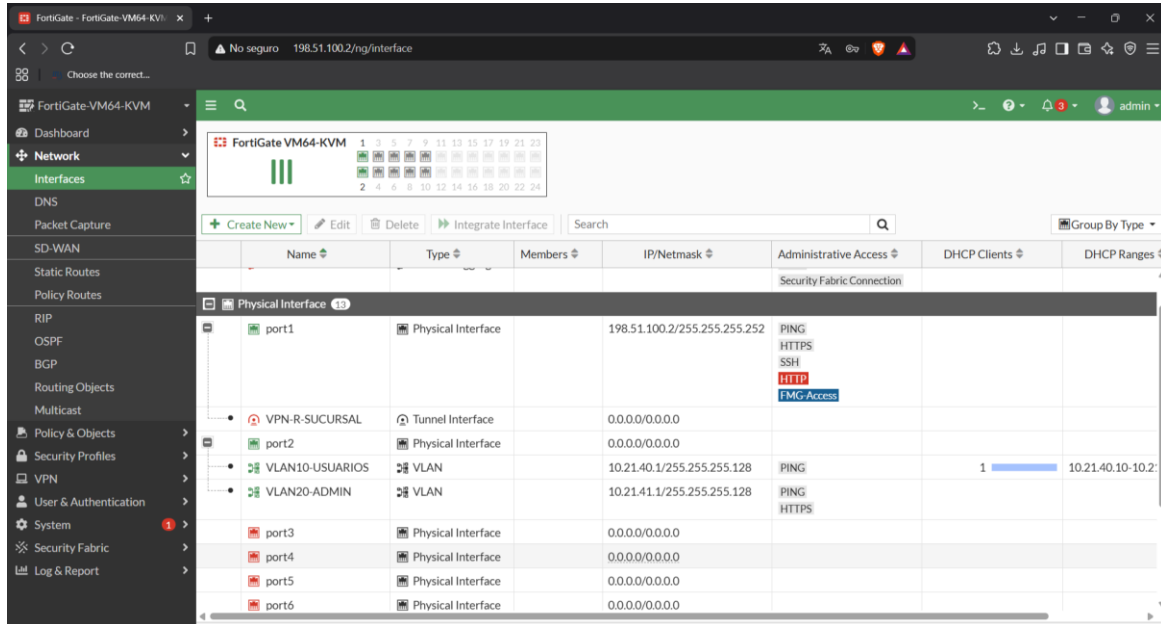


Evidencias de FortiGate Configuración mediante GUI

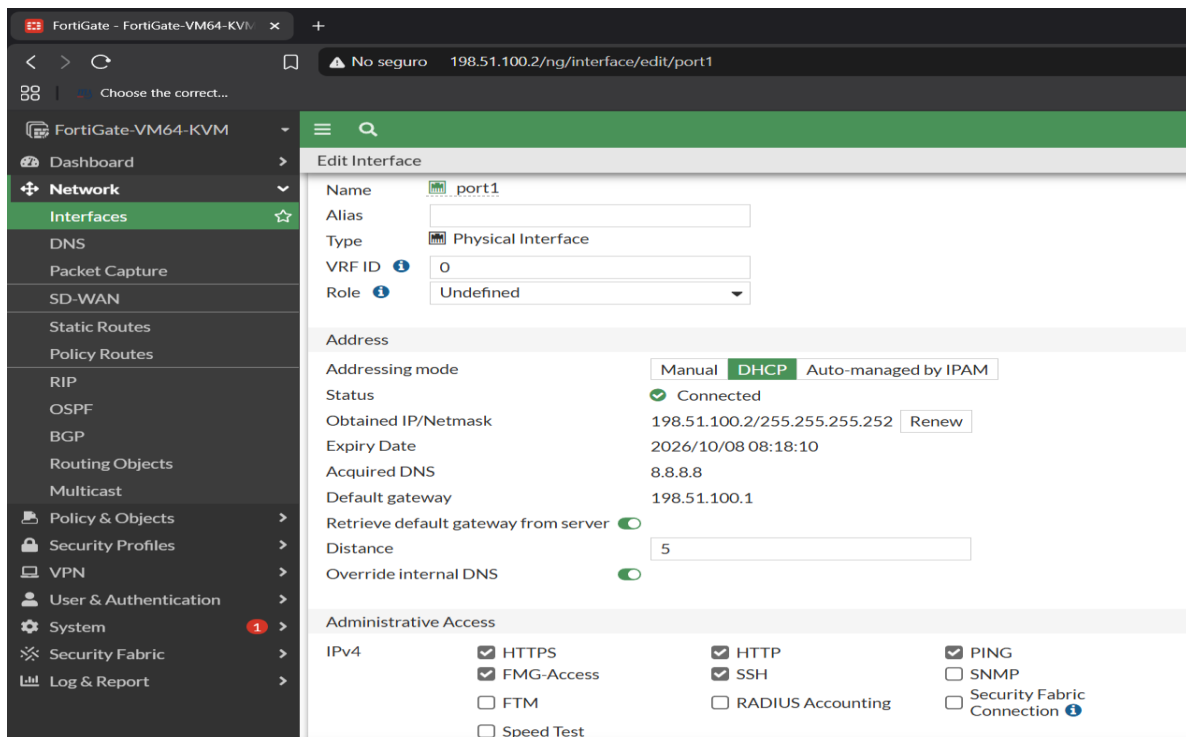
FortiGate - Vista general de interfaces

La vista general muestra port1 como enlace WAN, la interfaz de túnel VPN-R-SUCURSAL, port2 y las subinterfaces VLAN10-USUARIOS y VLAN20-ADMIN.



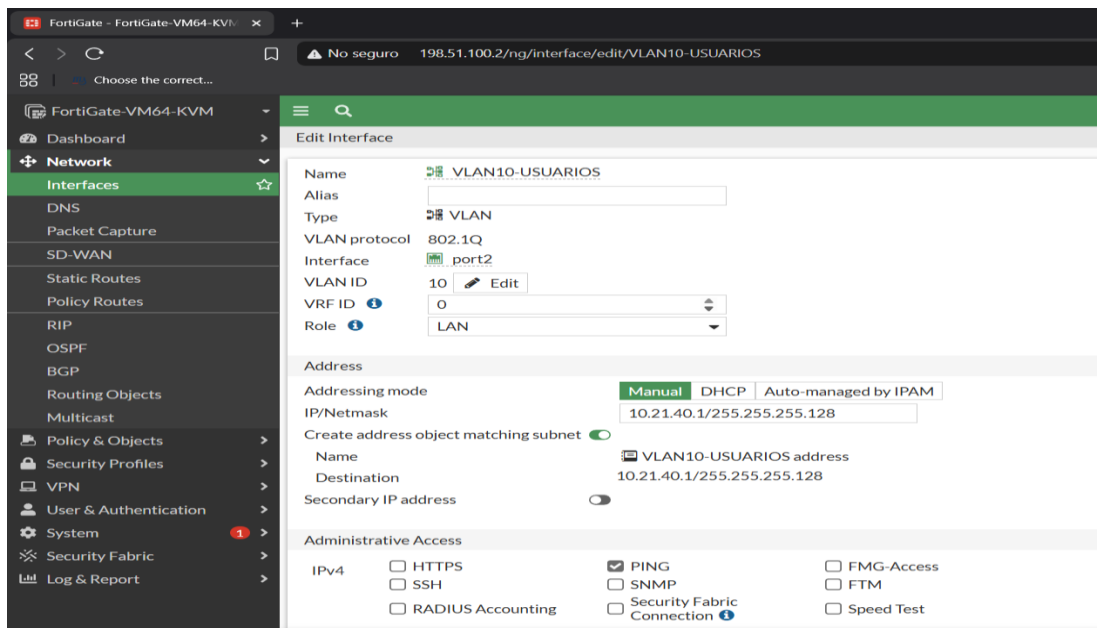
FortiGate - WAN (port1)

La interfaz WAN port1 utiliza DHCP y obtiene 198.51.100.2/30, gateway 198.51.100.1 y DNS 8.8.8.8. También mantiene acceso administrativo por HTTPS, HTTP, PING y SSH.



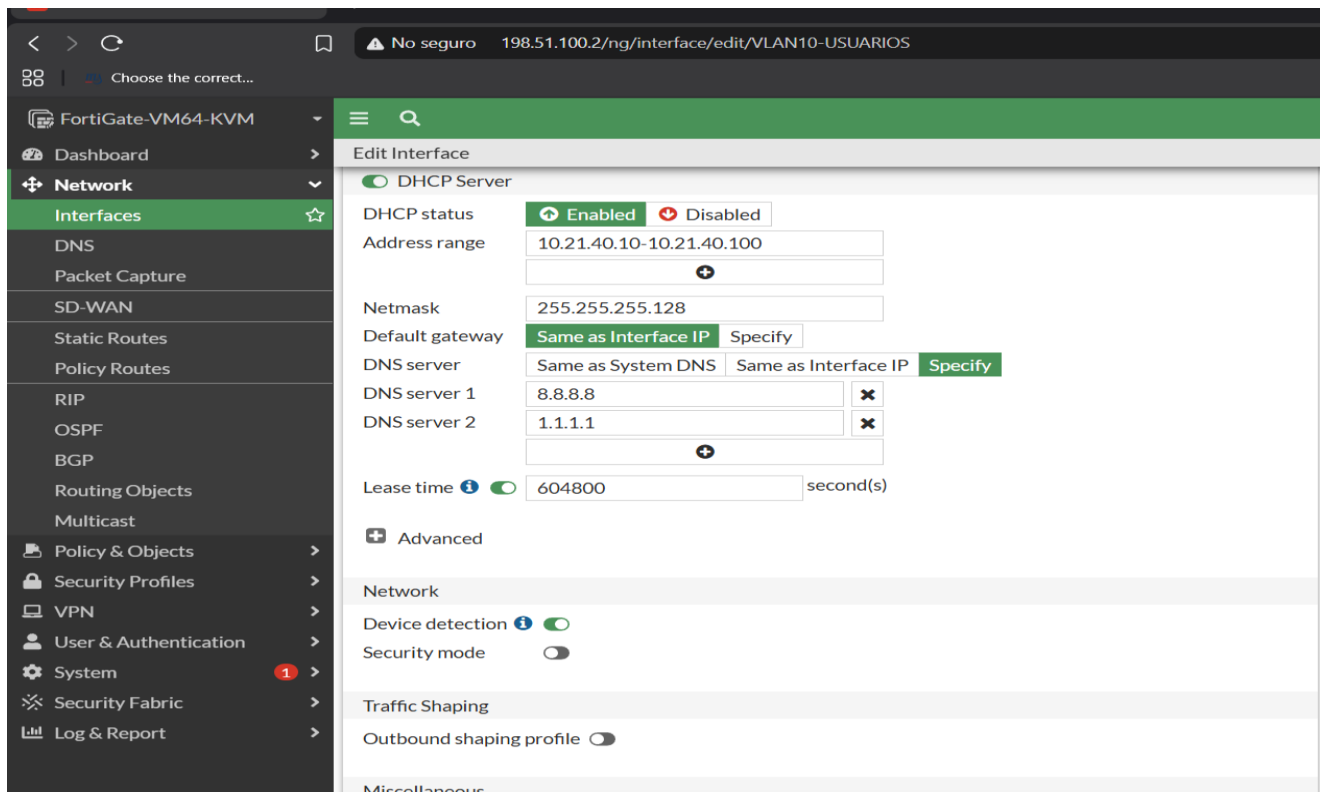
FortiGate - VLAN10-USUARIOS

La VLAN 10 está creada sobre port2 con encapsulación 802.1Q, VLAN ID 10 y dirección 10.21.40.1/25 para la red de usuarios.



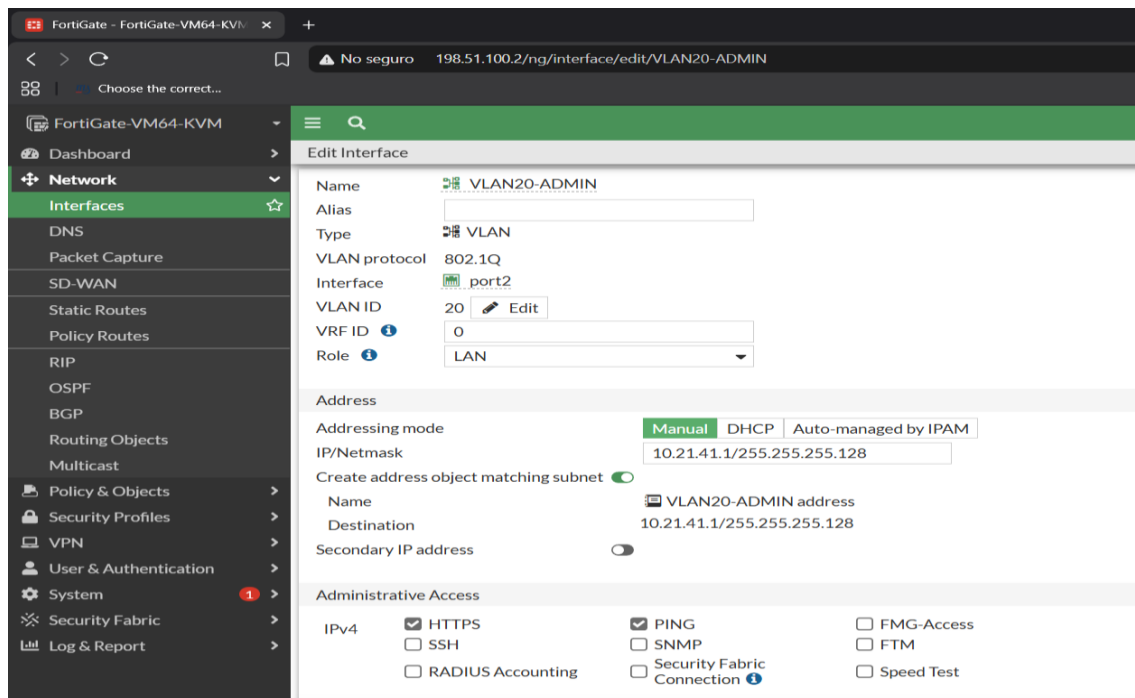
FortiGate - DHCP de VLAN 10

El servidor DHCP está habilitado con el rango 10.21.40.10 - 10.21.40.100, máscara /25, gateway de la interfaz y DNS 8.8.8.8 y 1.1.1.1.



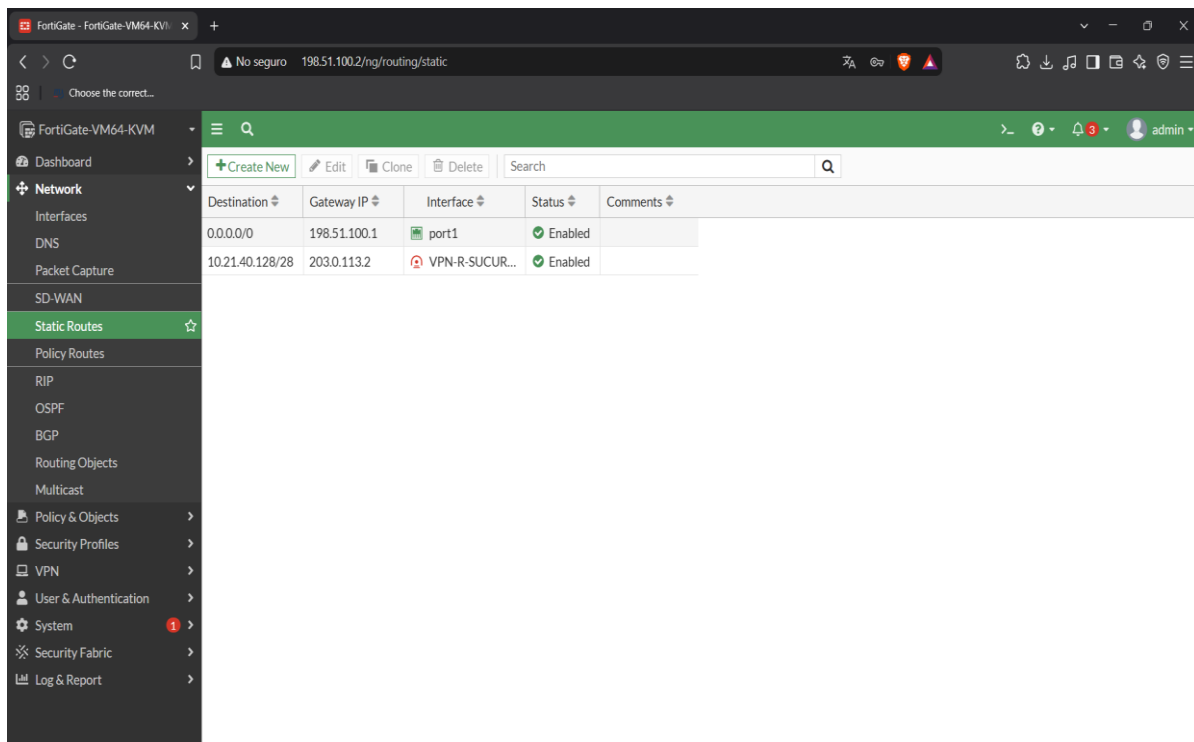
FortiGate - VLAN20-ADMIN

La VLAN 20 está creada sobre port2 con 802.1Q, VLAN ID 20 y dirección 10.21.41.1/25. El acceso administrativo habilitado incluye HTTPS y PING.



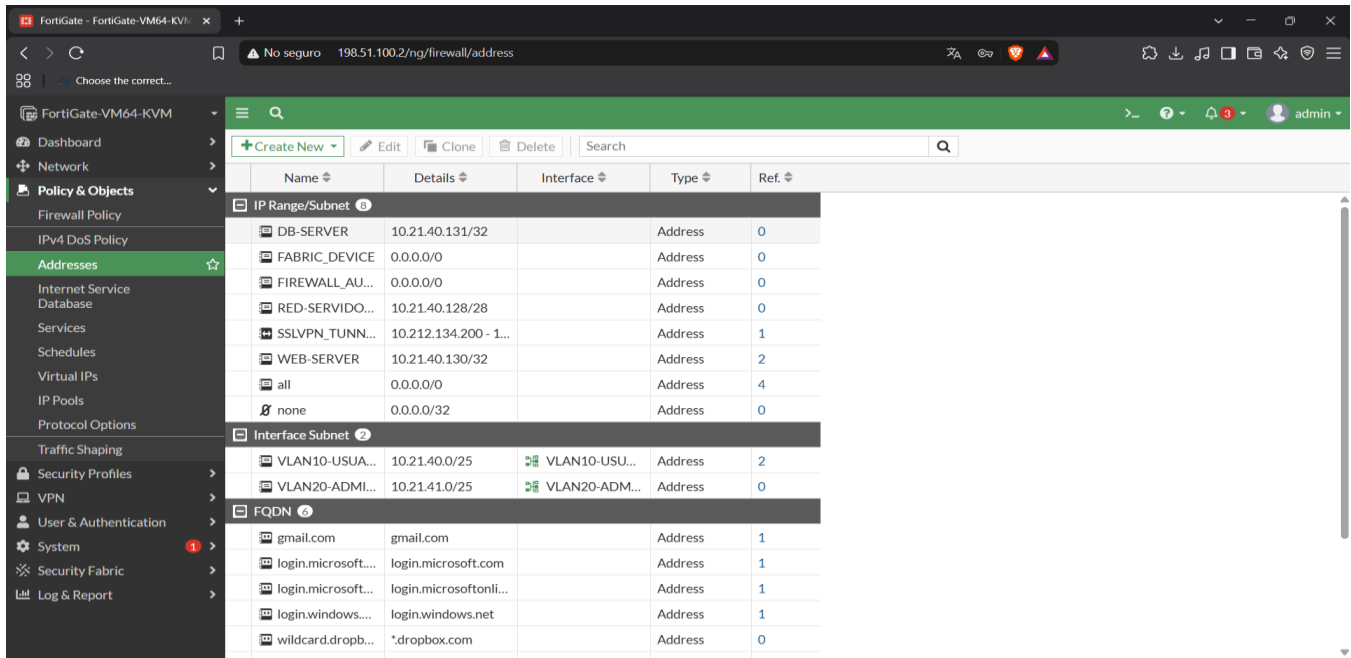
FortiGate - Rutas estáticas

Se configuró la ruta por defecto 0.0.0.0/0 hacia 198.51.100.1 por port1 y la ruta 10.21.40.128/28 hacia la sucursal mediante VPN-R-SUCURSAL.



FortiGate - Objetos de direcciones

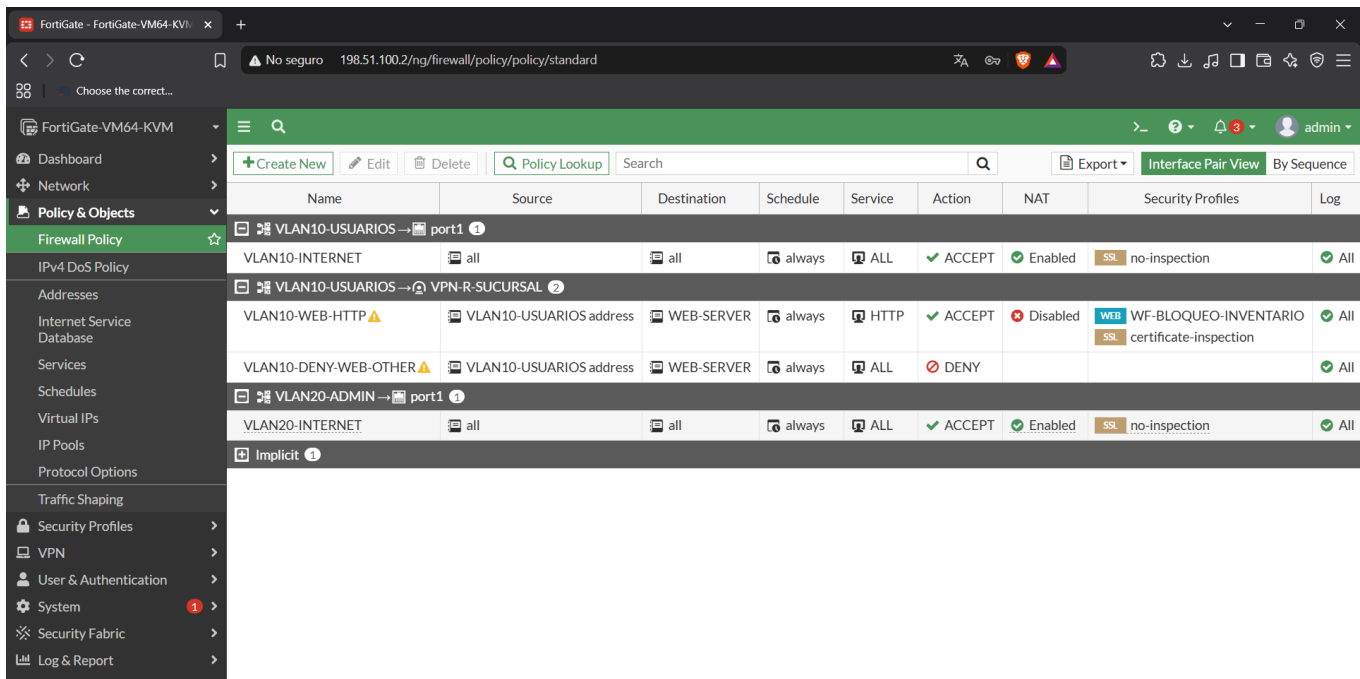
Se definieron objetos para WEB-SERVER, DB-SERVER, RED-SERVIDORES y las redes VLAN10-USUARIOS y VLAN20-ADMIN, utilizados por las políticas del firewall.



Name	Details	Interface	Type	Ref.
IP Range/Subnet 1				
DB-SERVER	10.21.40.131/32		Address	0
FABRIC_DEVICE	0.0.0.0/0		Address	0
FIREWALL_AU...	0.0.0.0/0		Address	0
RED-SERVIDO...	10.21.40.128/28		Address	0
SSLVPN_TUNN...	10.212.134.200 - 1...		Address	1
WEB-SERVER	10.21.40.130/32		Address	2
all	0.0.0.0/0		Address	4
none	0.0.0.0/32		Address	0
Interface Subnet 2				
VLAN10-USUA...	10.21.40.0/25	VLAN10-USU...	Address	2
VLAN20-ADMI...	10.21.41.0/25	VLAN20-ADM...	Address	0
FQDN 6				
gmail.com	gmail.com		Address	1
login.microsoft...	login.microsoft.com		Address	1
login.microsoft...	login.microsoftonli...		Address	1
login.windows...	login.windows.net		Address	1
wildcard.dropb...	*dropbox.com		Address	0

FortiGate - Políticas de firewall

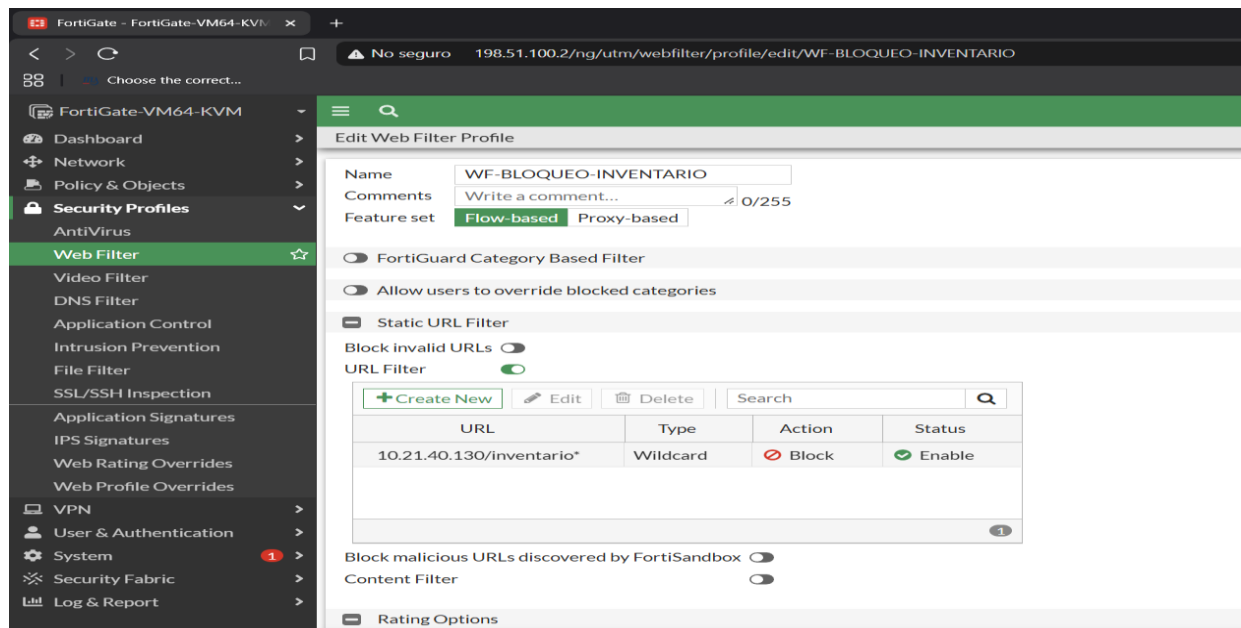
Las políticas permiten la salida a Internet de las VLAN y controlan el acceso de VLAN 10 al Web Server: HTTP permitido con Web Filter y el resto de servicios denegado y registrado.



Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log
VLAN10-USUARIOS -> port1 1								
VLAN10-INTERNET	all	all	always	ALL	ACCEPT	Enabled	no-inspection	All
VLAN10-USUARIOS -> VPN-R-SUCURSAL 2								
VLAN10-WEB-HTTP	VLAN10-USUARIOS address	WEB-SERVER	always	HTTP	ACCEPT	Disabled	WF-BLOQUEO-INVENTARIO certificate-inspection	All
VLAN10-DENY-WEB-OTHER	VLAN10-USUARIOS address	WEB-SERVER	always	ALL	DENY			All
VLAN20-ADMIN -> port1 1								
VLAN20-INTERNET	all	all	always	ALL	ACCEPT	Enabled	no-inspection	All
Implicit 1								

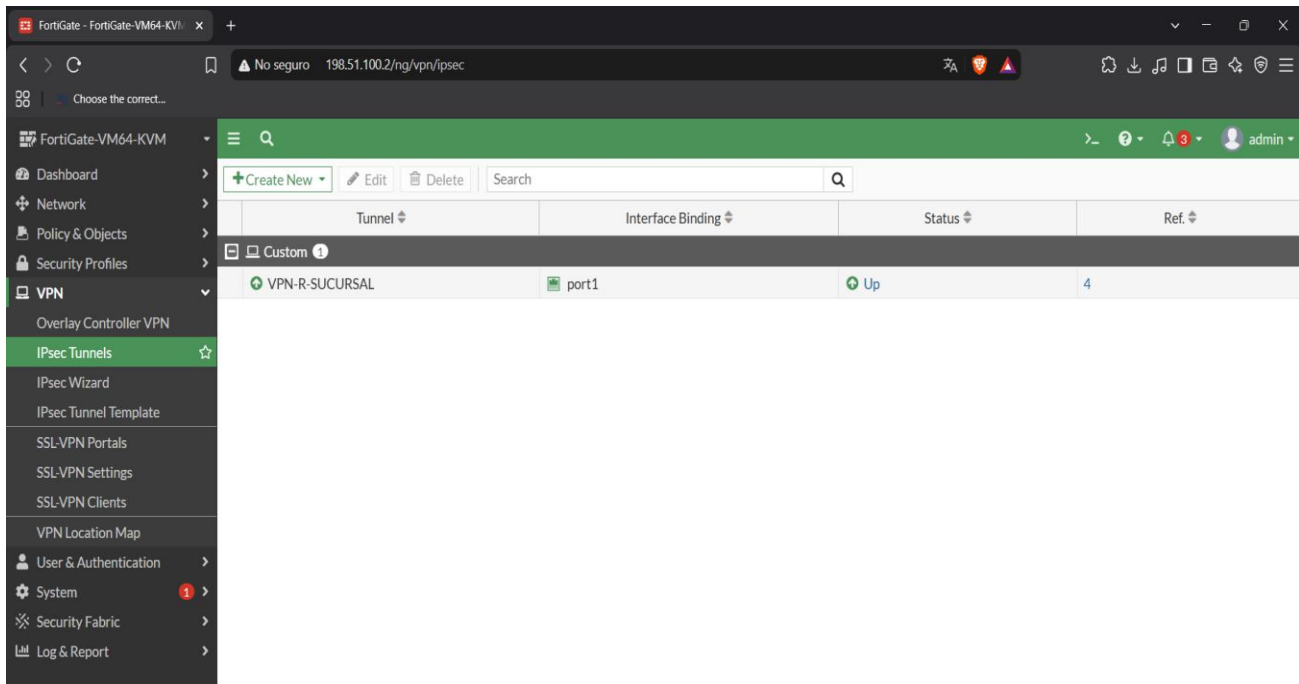
FortiGate - Web Filter

El perfil WF-BLOQUEO-INVENTARIO utiliza Static URL Filter en modo Flow-based para bloquear 10.21.40.130/inventario* mediante coincidencia Wildcard y acción Block.



FortiGate - Estado del túnel IPsec

El túnel VPN-R-SUCURSAL se encuentra asociado a port1 y aparece en estado Up, confirmando que la VPN Site-to-Site está establecida.



FortiGate - VPN IPsec Phase 1

La Fase 1 utiliza peer remoto 203.0.113.2, port1, autenticación Pre-shared Key, IKEv1 en modo Main, DES/SHA1, Diffie-Hellman Group 5 y lifetime de 86400 segundos.

The screenshot shows the FortiGate web interface for editing a VPN tunnel named 'VPN-R-SUCURSAL'. The left sidebar contains a navigation menu with options like Dashboard, Network, Policy & Objects, Security Profiles, VPN, and System. The main content area is titled 'Edit VPN Tunnel' and is divided into several sections:

- Network:** Remote Gateway : Static IP Address (203.0.113.2) , Interface : port1
- Authentication:** Authentication Method : Pre-shared Key, IKE Version : 1 , Mode : Main (ID protection)
- Phase 1 Proposal:** A yellow highlighted section containing:
 - Encryption: DES
 - Authentication: SHA1
 - Diffie-Hellman Group: A grid of radio buttons where group 5 is selected.
 - Key Lifetime (seconds): 86400
 - Local ID: (empty field)
- XAUTH:** Type : Disabled
- Phase 2 Selectors:** A table listing selectors for Phase 2.

Name	Local Address	Remote Address
VPN-R-SUCURSAL	10.21.40.0/255.255.255.128	10.21.40.128/255.255.255.240

FortiGate - VPN IPsec Phase 2

La Fase 2 protege 10.21.40.0/25 hacia 10.21.40.128/28 con DES/SHA1, Replay Detection, PFS habilitado y Diffie-Hellman Group 5.

The screenshot shows the 'Edit Phase 2' configuration for the 'VPN-R-SUCURSAL' tunnel. The interface includes a table for Phase 2 selectors and a detailed configuration section for the selected selector.

Name	Local Address	Remote Address
VPN-R-SUCURSAL	10.21.40.0/255.255.255.128	10.21.40.128/255.255.255.240

Edit Phase 2

Name: VPN-R-SUCURSAL

Comments: (empty field)

Local Address: Subnet 10.21.40.0/255.255.255

Remote Address: Subnet 10.21.40.128/255.255.2

Advanced...

Phase 2 Proposal:

- Encryption: DES
- Authentication: SHA1
- Enable Replay Detection: ☒
- Enable Perfect Forward Secrecy (PFS): ☒
- Diffie-Hellman Group: A grid of radio buttons where group 5 is selected.
- Local Port: All ☒
- Remote Port: All ☒
- Protocol: All ☒
- Auto-negotiate: ☐